



# Analytics & Big Data

Session 3: Analytical data architecture

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(2026-03-30)

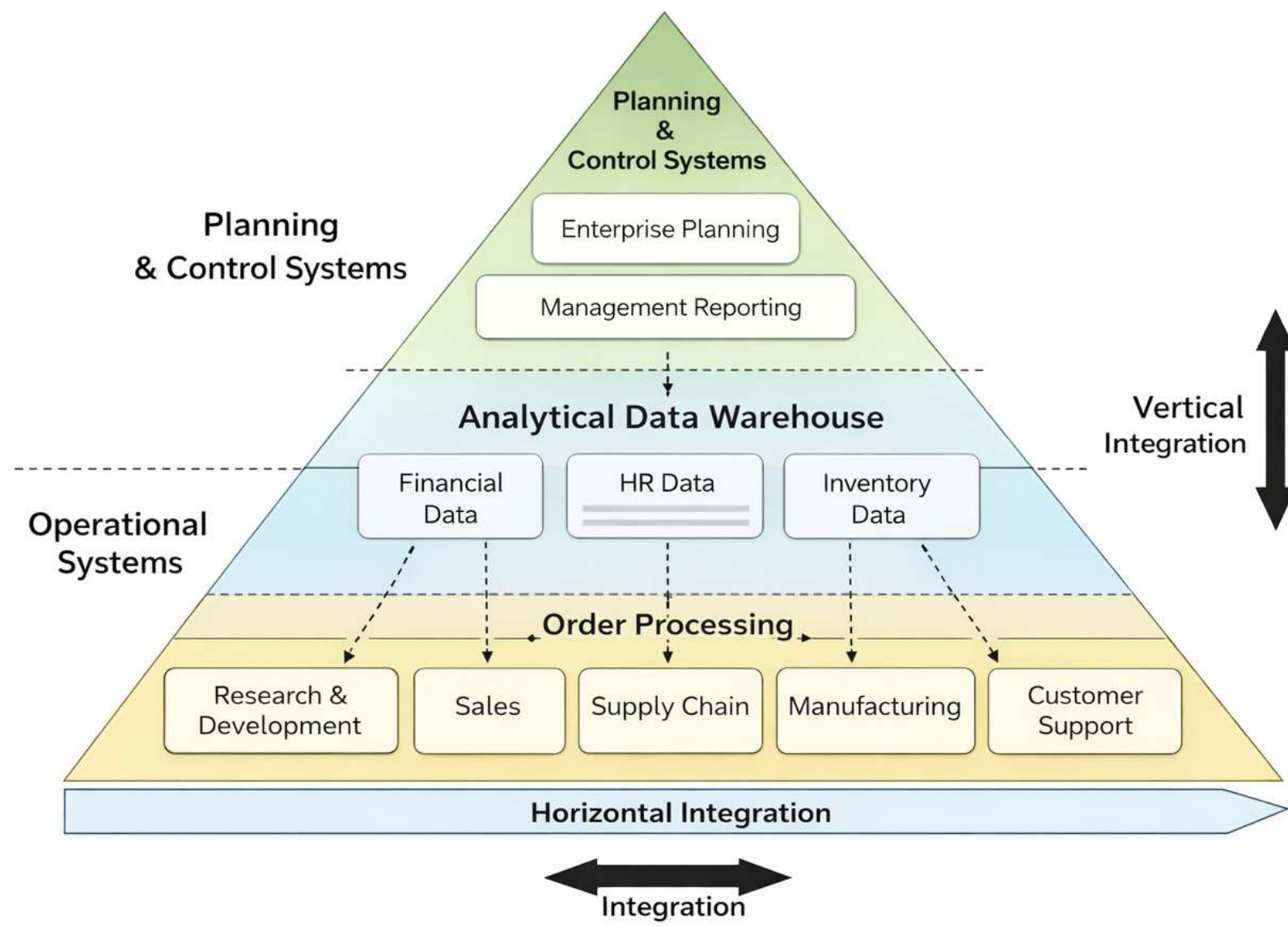
## Learning objectives

- **Outline** the architecture of data warehouses.
- **Explain** dimensional modeling concepts and common schemata.
- **Design** fact and dimension schemata from business questions and operational data.

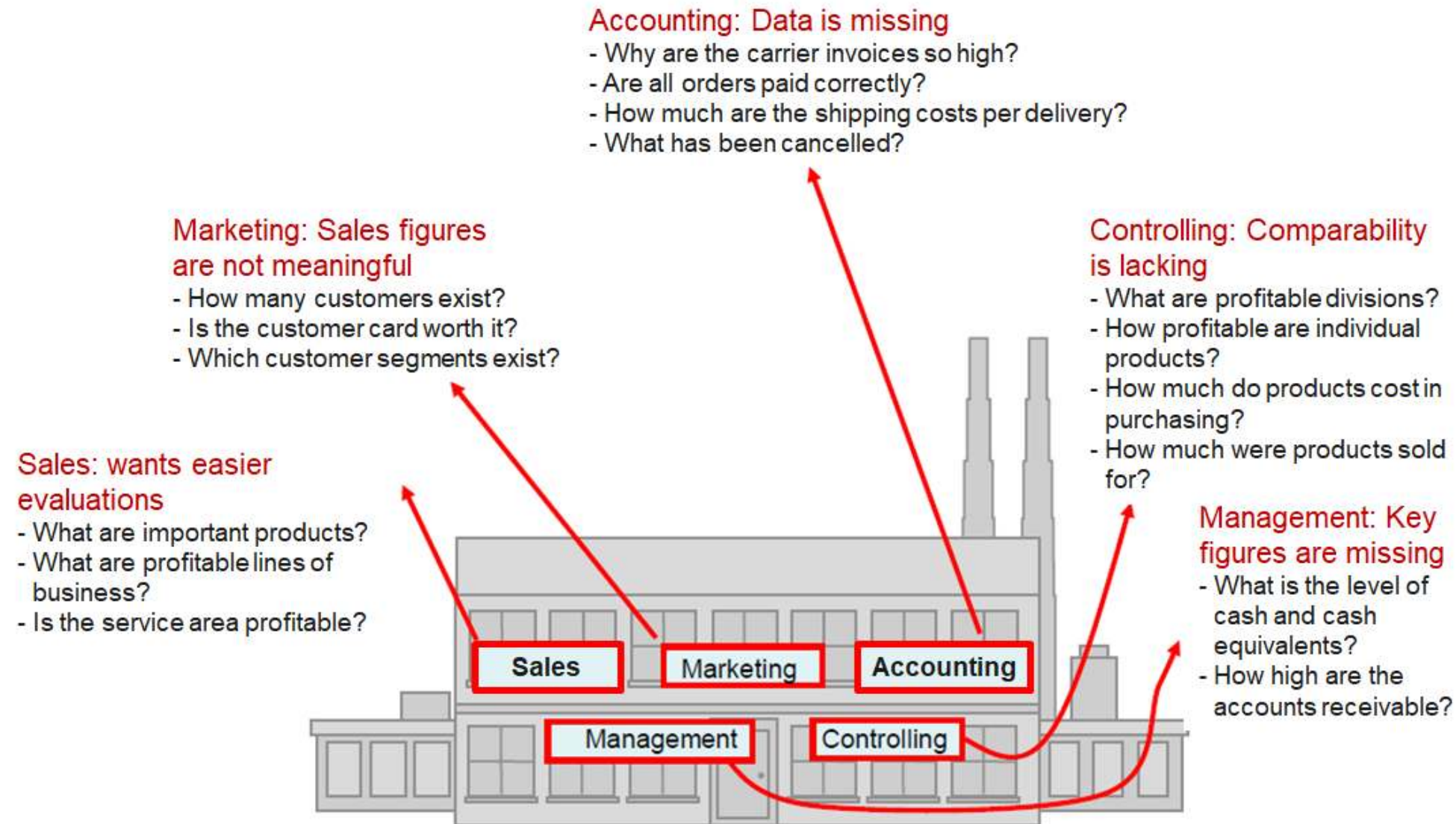


# Data warehouse

# Application systems pyramid



# Typical descriptive questions



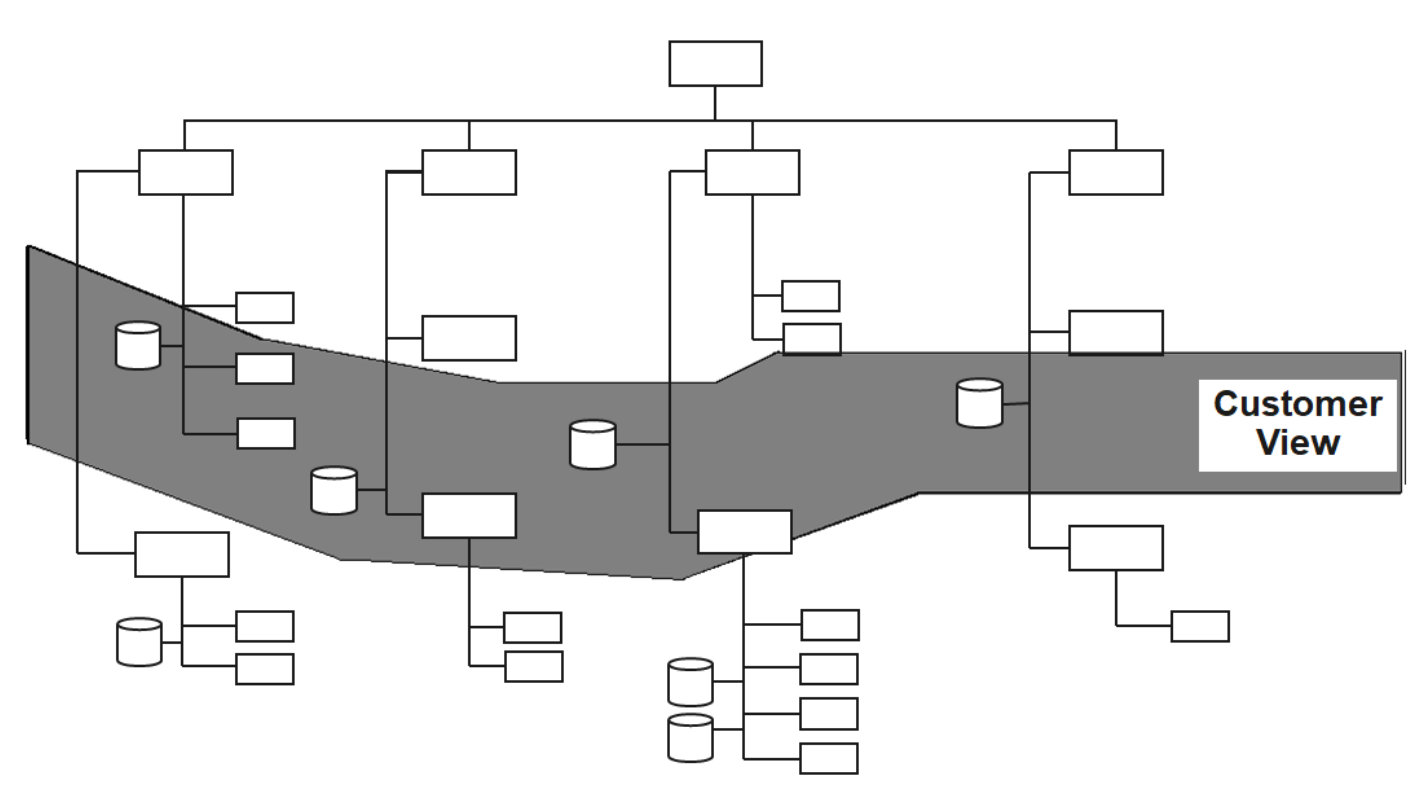
— Source: Alfred Schlaucher, Oracle, p. 15

# Characteristics of operational and analytical databases



|                        | Operational database  | Analytical database   |
|------------------------|-----------------------|-----------------------|
| Target group           | Operational employees | Management, Analysts  |
| Access frequency       | High                  | Slow to medium        |
| Data volume per access | Low to medium volume  | High volume           |
| Required response time | Very short            | Short to medium       |
| Level of data          | Detailed              | Aggregated, processed |
| Queries                | Predictable, periodic | Unpredictable, ad hoc |
| Focal period of time   | Current               | Past to future        |
| Time horizon           | 1–3 months            | Years or decades      |

# Contradiction between operational data architecture and analytical information needs



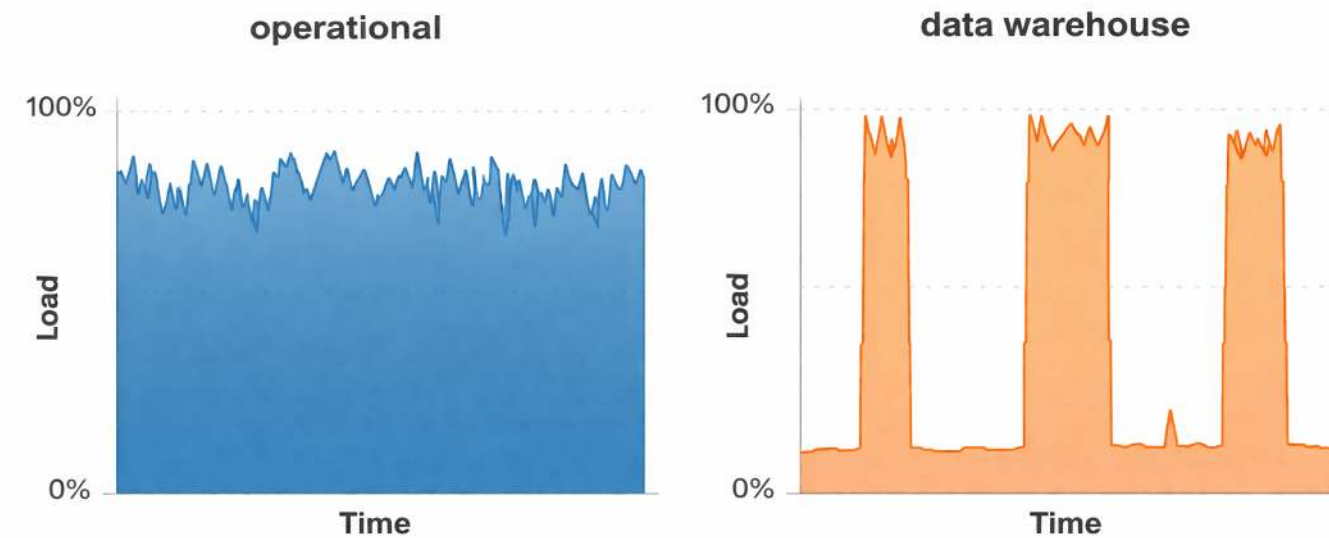
Organizations structure operational data by departments and processes, but analytical questions require cross-departmental views.

— Source: Jung/Winter (2000): Data Warehousing Strategie: Erfahrungen, Methoden, Visionen, Berlin, p. 7.

# Different intensities of use



The hardware load differs in operational application systems and data warehouse systems over time



=> A data warehouse relieves the operational systems

— Source: Inmon ([2005](#)), p. 23.

# Definition: Data warehouse

The term Data Warehouse was coined by Inmon ([1995](#)), who defined it as follows:

“A data warehouse is a subject-oriented, integrated, time-variant and non-volatile collection of data in support of management’s decision-making process.”

He defined the terms in the sentence as follows:

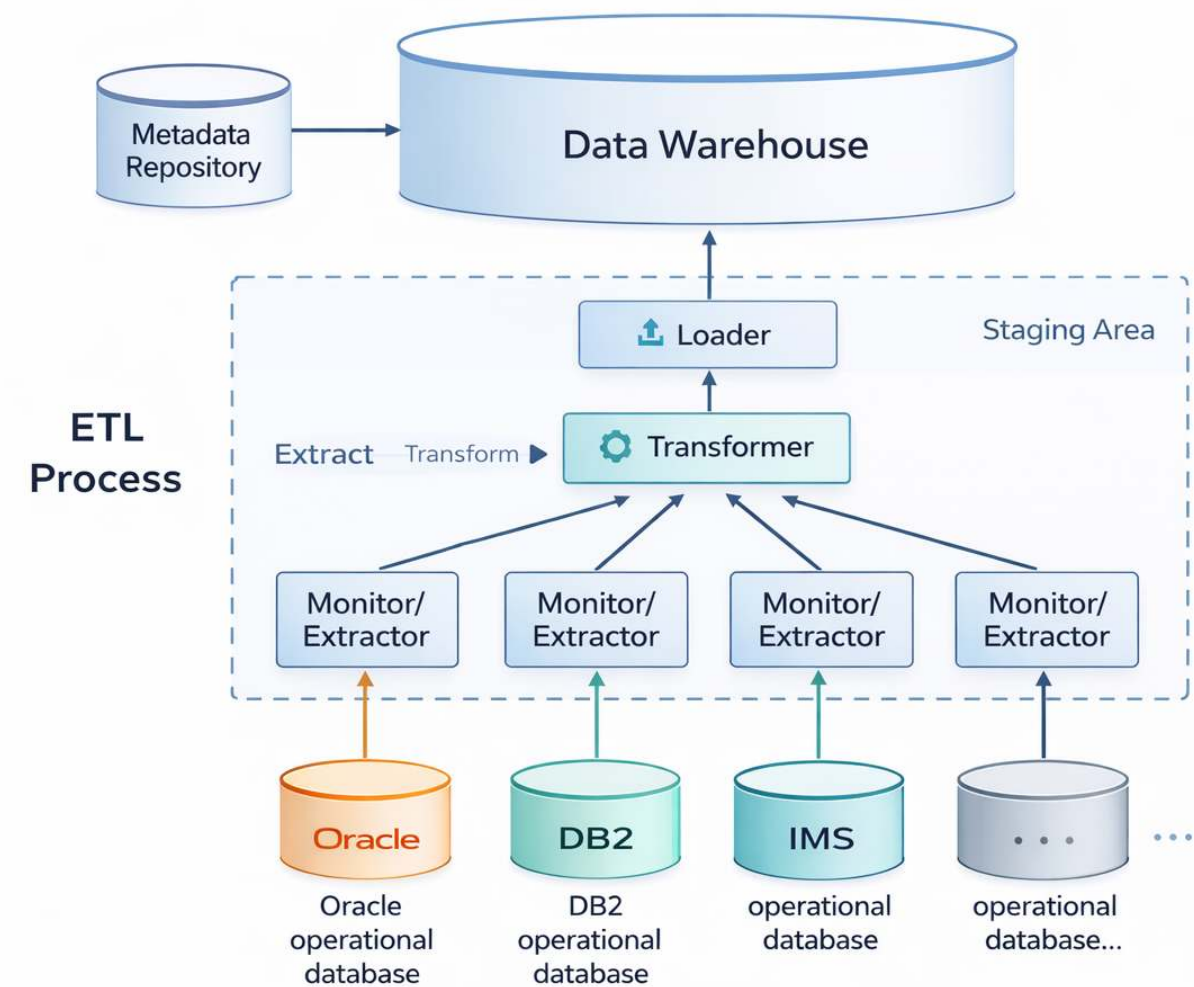
**Subject-oriented:** Data that gives information about a particular subject instead of about a company’s ongoing operations. Rather than reflecting fragmented departmental systems, it integrates data into consistent, cross-departmental subject areas.

**Integrated:** Data that is gathered into the data warehouse from a variety of sources and merged into a coherent whole.

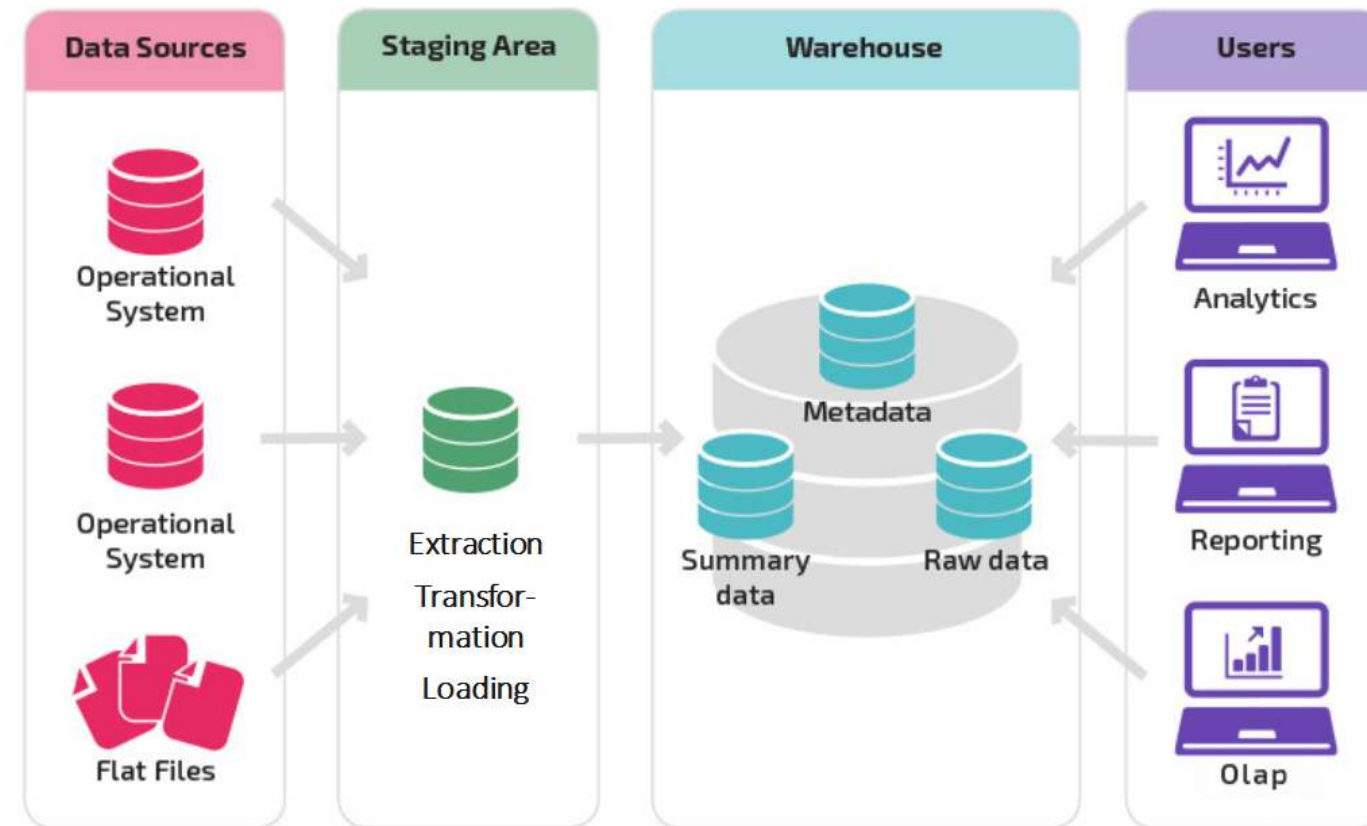
**Time-variant:** All data in the data warehouse is identified with a particular time period.

**Non-volatile:** Data is stable in a data warehouse. More data is added but data is never removed. This enables management to gain a consistent picture of the business.

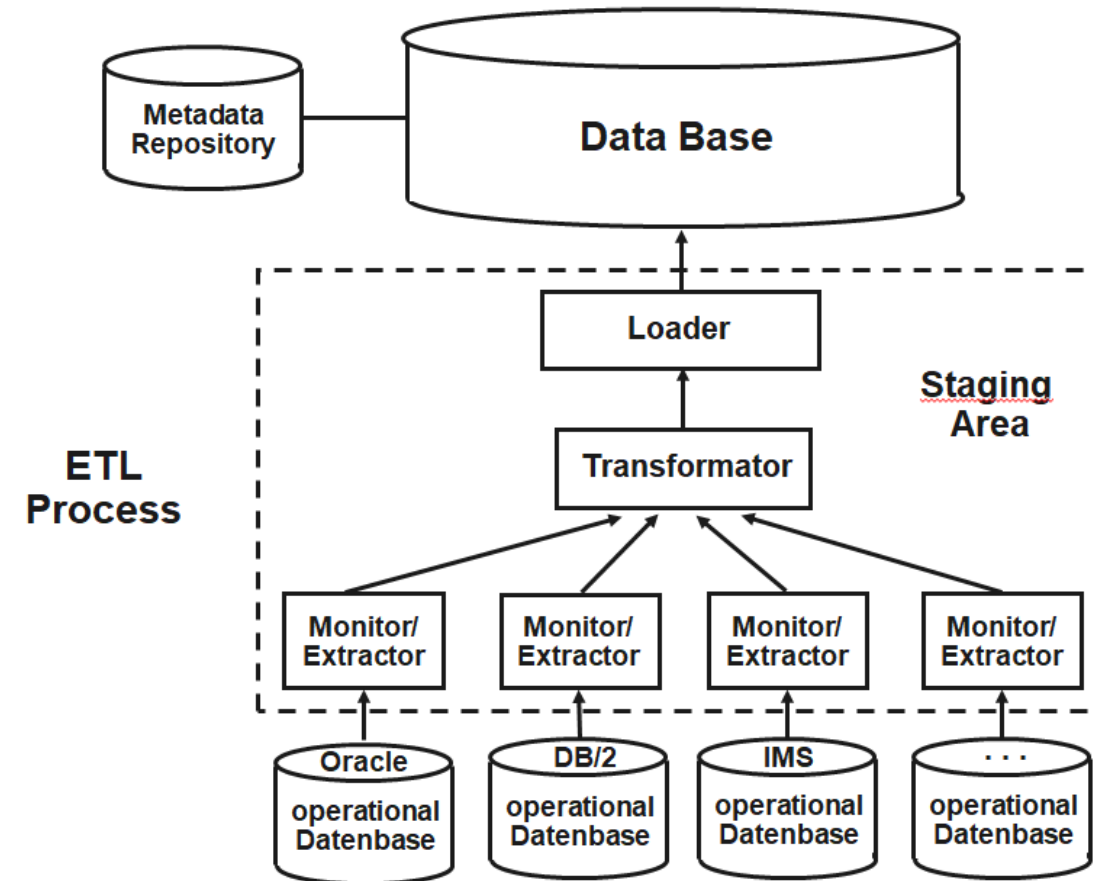
# The data warehouse concept



# Layers of a data warehouse system



# Components of a data warehouse



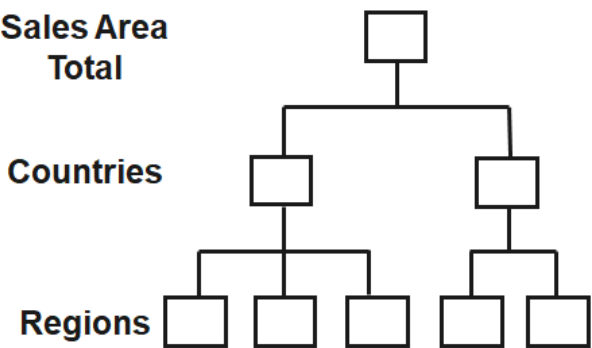
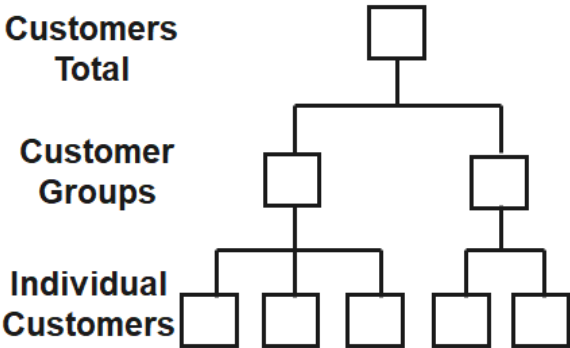
# Structure of the database

The database represents the core of the data warehouse. It contains current and historical data from all relevant areas of the company. In contrast to the operational databases, the data here is not structured according to the business processes and functions of the company, but is aligned with the facts that determine the company. They are organized in such a way that they can be viewed from different dimensions.

Frequently used dimensions are:

- Product structure (e.g., product groups and individual products)
- Regional structure (e.g., region, country, continent)
- Customer structure (e.g., customer groups or segments, and individual customers)
- Time structure (e.g., month, quarter, year)
- Business parameters (e.g., sales, contribution margin, profit)
- Characteristics (e.g., target, actual, deviations)

# Examples of aggregation levels in a data warehouse





# ETL pipeline

The data in the data warehouse can normally only be accessed in a **read-only manner**. The only exception to this are ETL tools, which are responsible for transferring data from internal and external information sources. They automate the process of data acquisition from these sources.

The timeliness with which the transfer to the database is performed can be defined flexibly, e.g., within defined time intervals, immediately after each change, or at the explicit request of the user.

ETL programs include:

- **Monitors:** Detect and report changes in individual sources that are relevant to the data warehouse.
- **Extractors:** Select and transport data from the data sources into the workspace.
- **Transformers:** Unify, clean, integrate, consolidate, aggregate, and augment extracted data in the workspace.
- **Loaders:** Load transformed data from the workspace into the data warehouse after the data retrieval process is complete.

# Staging area



The staging area is a temporary cache that performs the ETL process. It includes the extraction, transformation and loading of the data into the data warehouse.

Neither the operational systems nor the data warehouse are affected by the process. The data can be transferred incrementally from the operational systems without having to have a permanent connection to them. It is stored temporarily and only transformed when it is sufficiently complete.

It is then loaded en bloc into the data warehouse, so that no inconsistencies can occur here either.

If the ETL process is successful, the data is deleted from the staging area.

# The metadata repository

Metadata supports data management and serves as a descriptor for an object that holds some data or information. In data warehouses, it is collectively organized in a catalog called the metadata repository.

Metadata is classified into three types:

- **Technical metadata:** Provides information about the structure of data, where it resides, and other technical details related to locating data in its native database.
- **Business metadata:** Describes the actual data in business-understandable terms. It can provide insights into the type of data, its origin, definition, quality, and relationships with other entities in the data warehouse.
- **Process metadata:** Stores information about the occurrence and outcomes of operations taking place in the data warehouse.

# Data warehouse and data mart

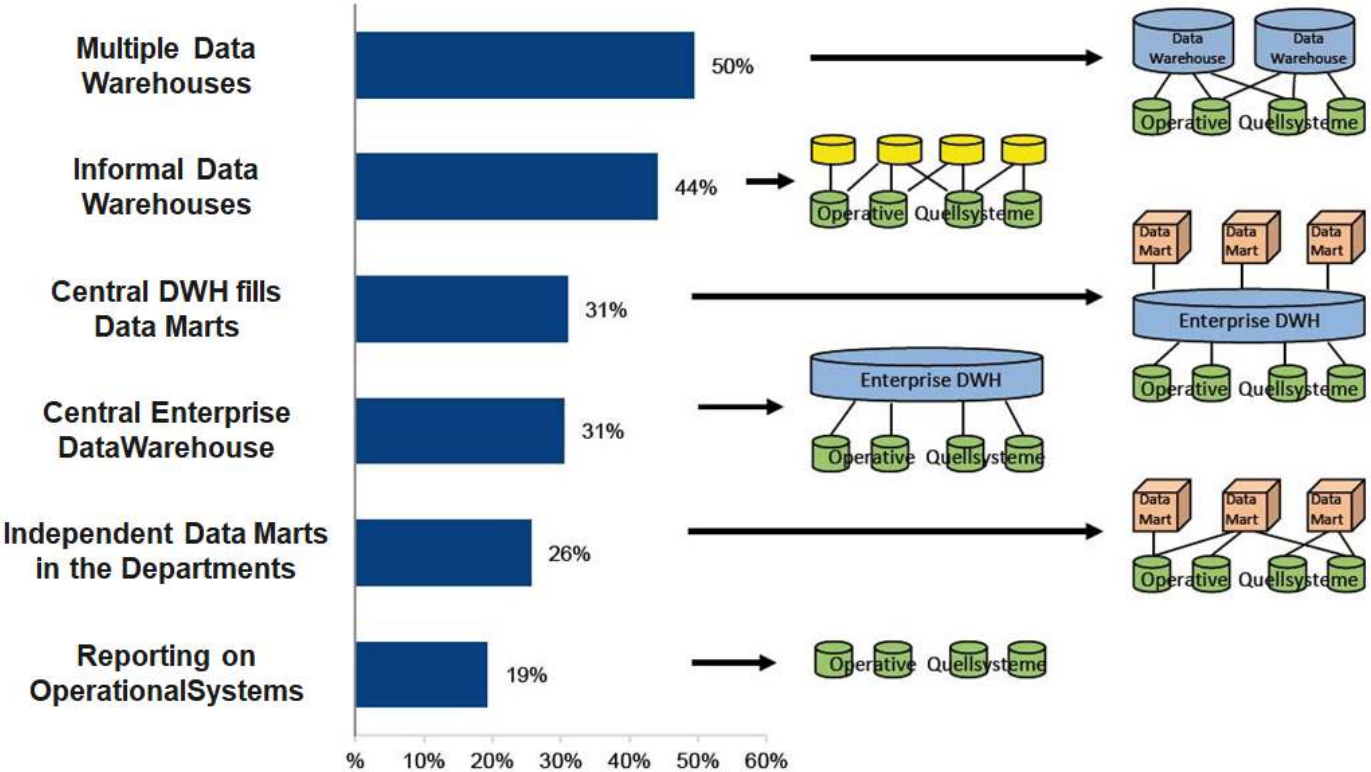


A single-subject data warehouse is typically referred to as a Data Mart, while Data Warehouses are generally enterprise in scope.

Comparison: Data warehouse and data mart

|              | Data mart           | Data warehouse      |
|--------------|---------------------|---------------------|
| Application  | Division/Department | Company             |
| Data design  | Optimized           | Generalist          |
| Data volume  | Gigabyte scale      | Terabyte scale      |
| Origin       | Task oriented       | Data model oriented |
| Requirements | Specific            | Versatile           |

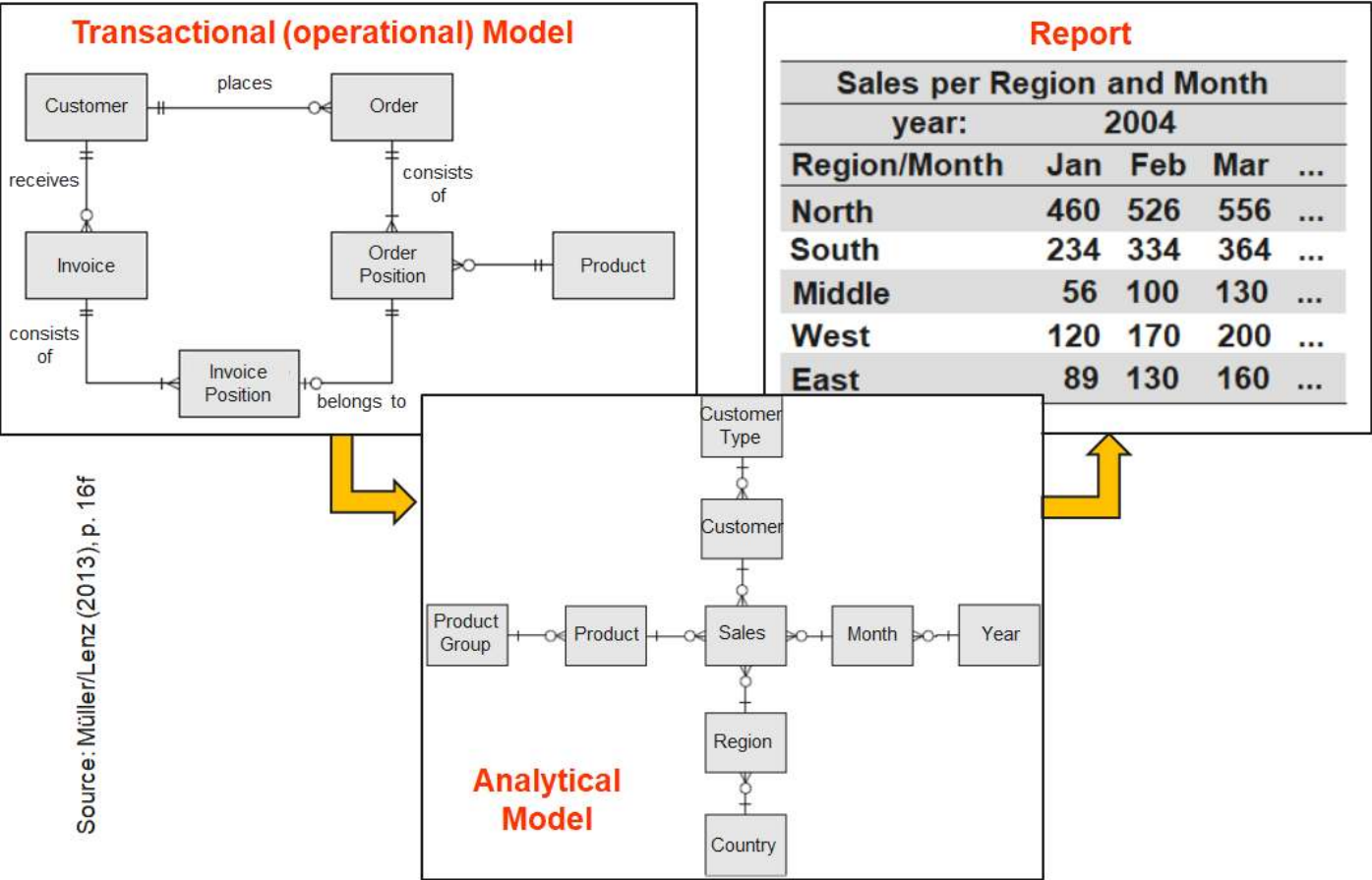
# Architectural variants of data warehouses in practice





# Dimensional modeling

# From transactions to reports



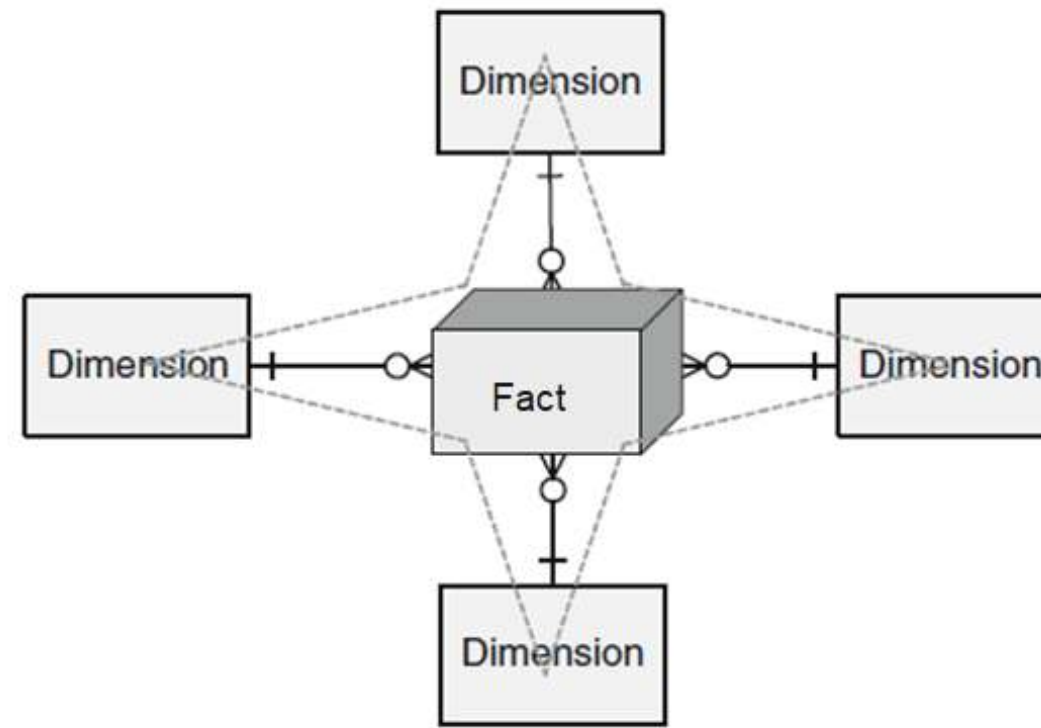
Source: Müller/Lenz (2013), p. 16f



# Star schema (I)

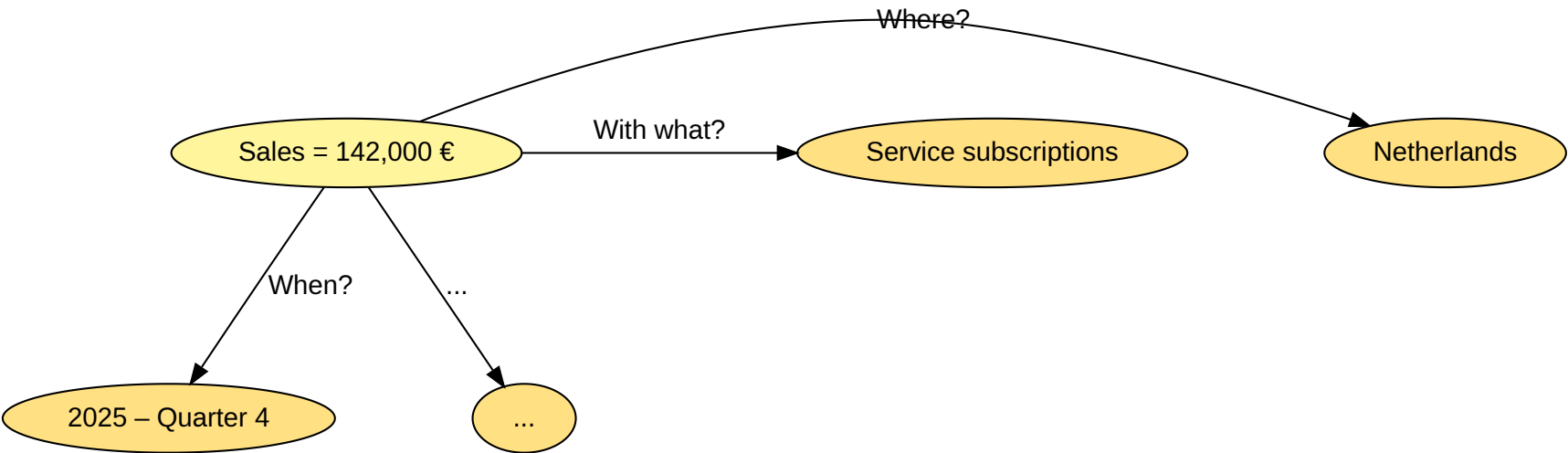


The Star schema has a central fact table and exactly one dimension table for each dimension.

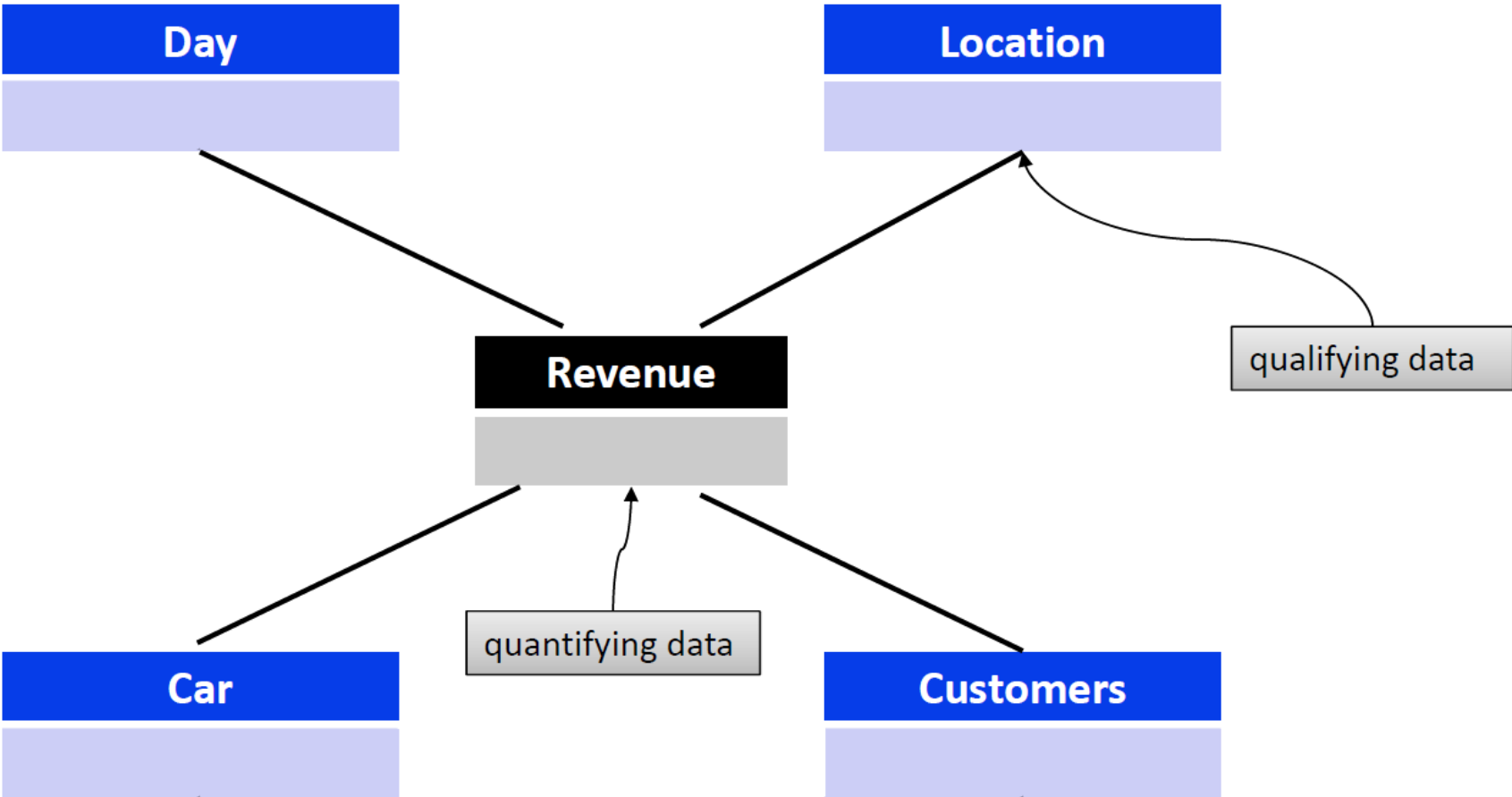


The cardinality between the fact table and a dimension table is  $n:1$ , i.e. a fact (e.g. sales: 67,000 euros) is described by one dimension expression for each dimension (e.g. time: 1st quarter, region: Hesse, product group: books). A dimension characteristic (e.g. product group: books) is associated with 0, 1 or  $n$  facts.

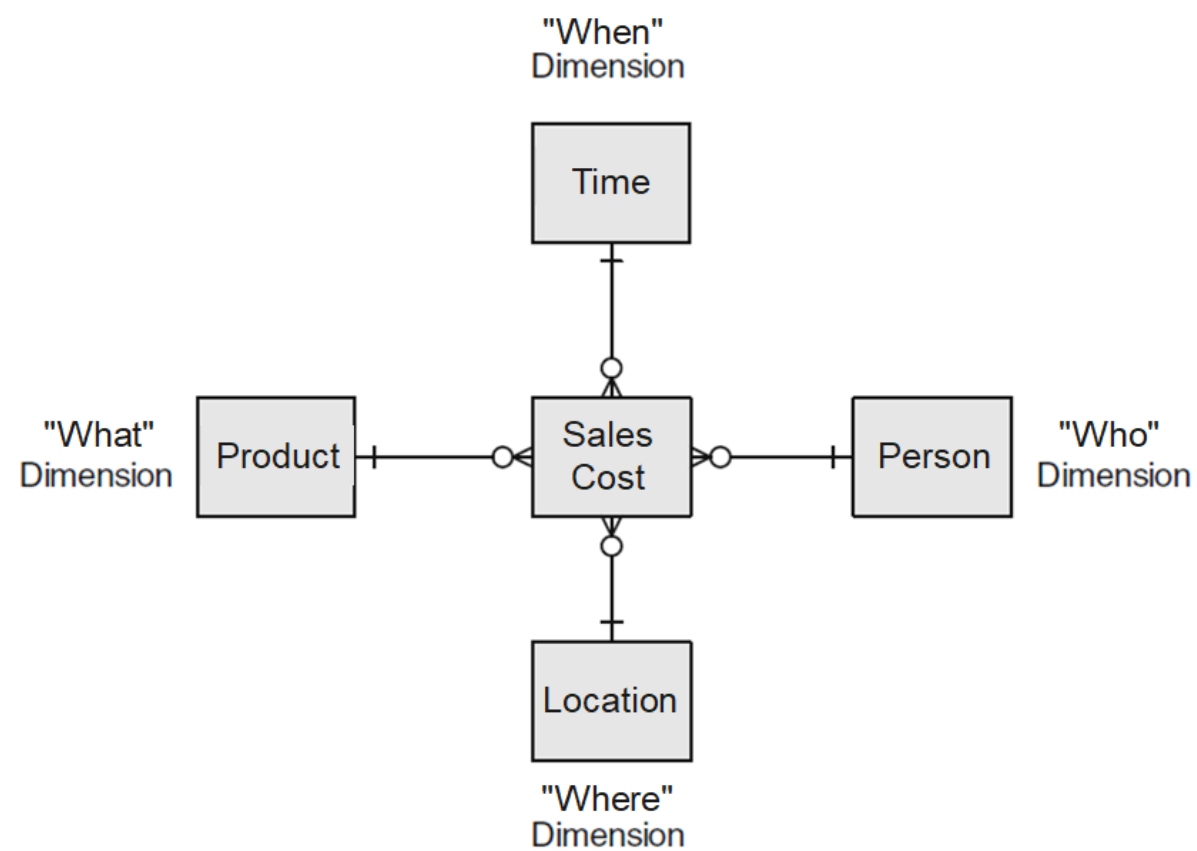
# Star schema (II)



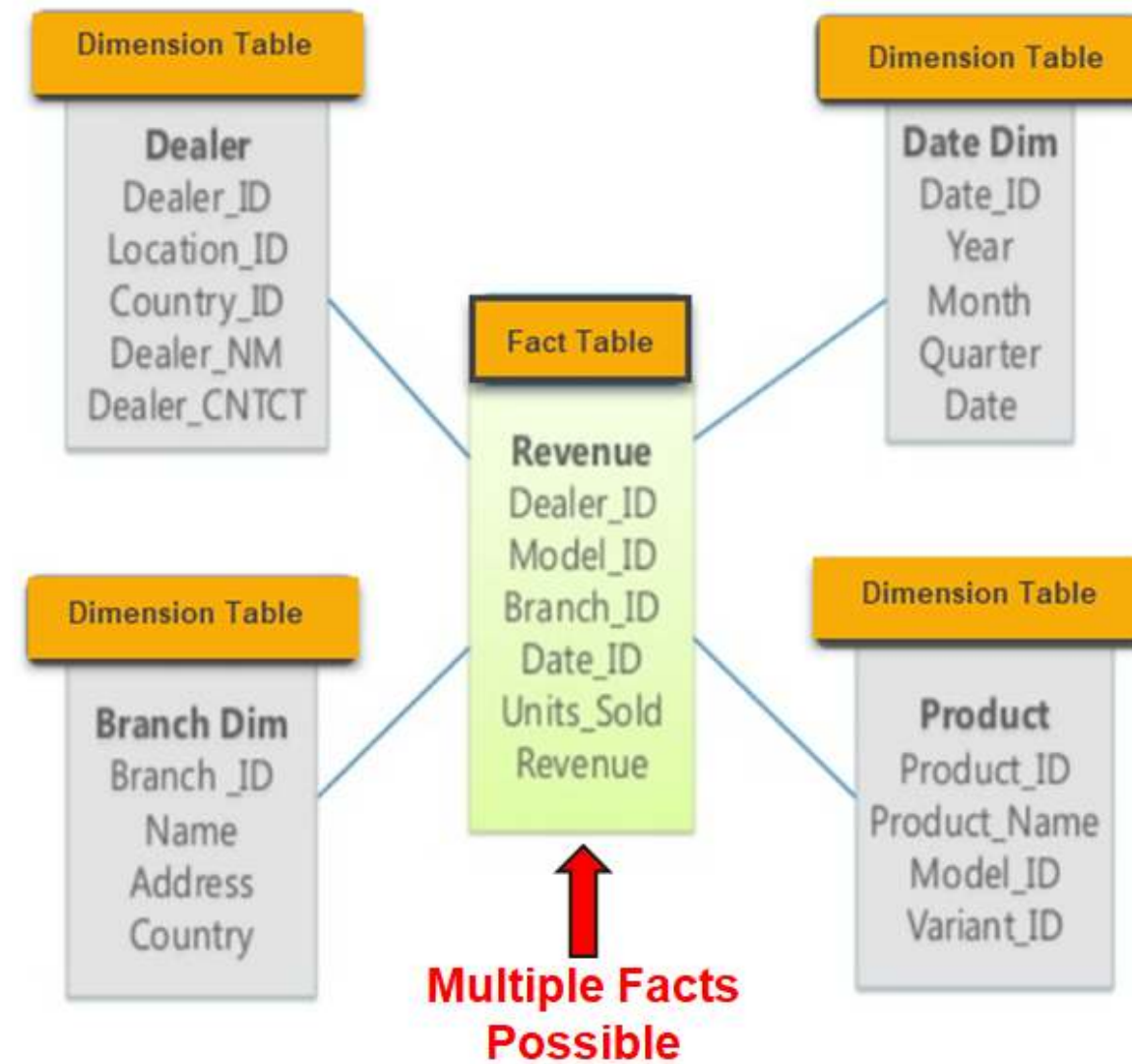
# Star schema (III)



# Star schema table design (I)

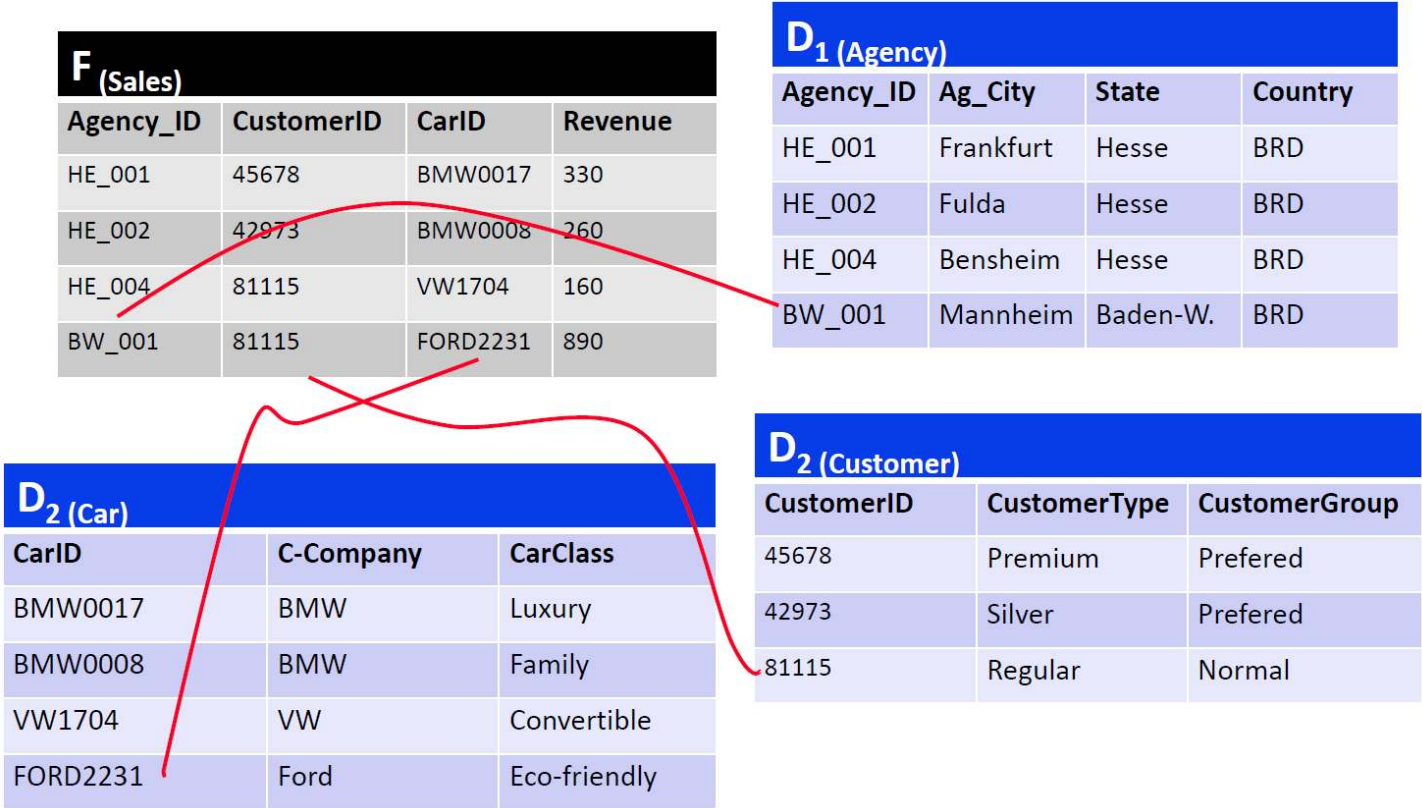


# Star schema table design (II)



— Source: <https://www.guru99.com/star-snowflake-data-warehousing.html>

# Star schema example



# Queries on the star schema

Example:

In which years were the most cars purchased by female customers in Hesse in the 1st quarter?

```

1 select z.Year as Year, sum(v.Amount) as Total_Amount
2 from Branch f, Product p, Time z, Customer k, Sales v
3 where z.Quarter = 1 and k.gender = 'f' and p.Product_type = 'Car' and f.state = 'Hesse' \
4       and v.Date = z.Date and v.ProductID = p.ProductID and v.Branch = f.Branchname \
5       and v.CustomerID = k.CustomerID
6 group by z.Year
7 order by Total_Amount Descending;

```

| Year | Total amount |
|------|--------------|
| 2004 | 745          |
| 2005 | 710          |
| 2003 | 650          |

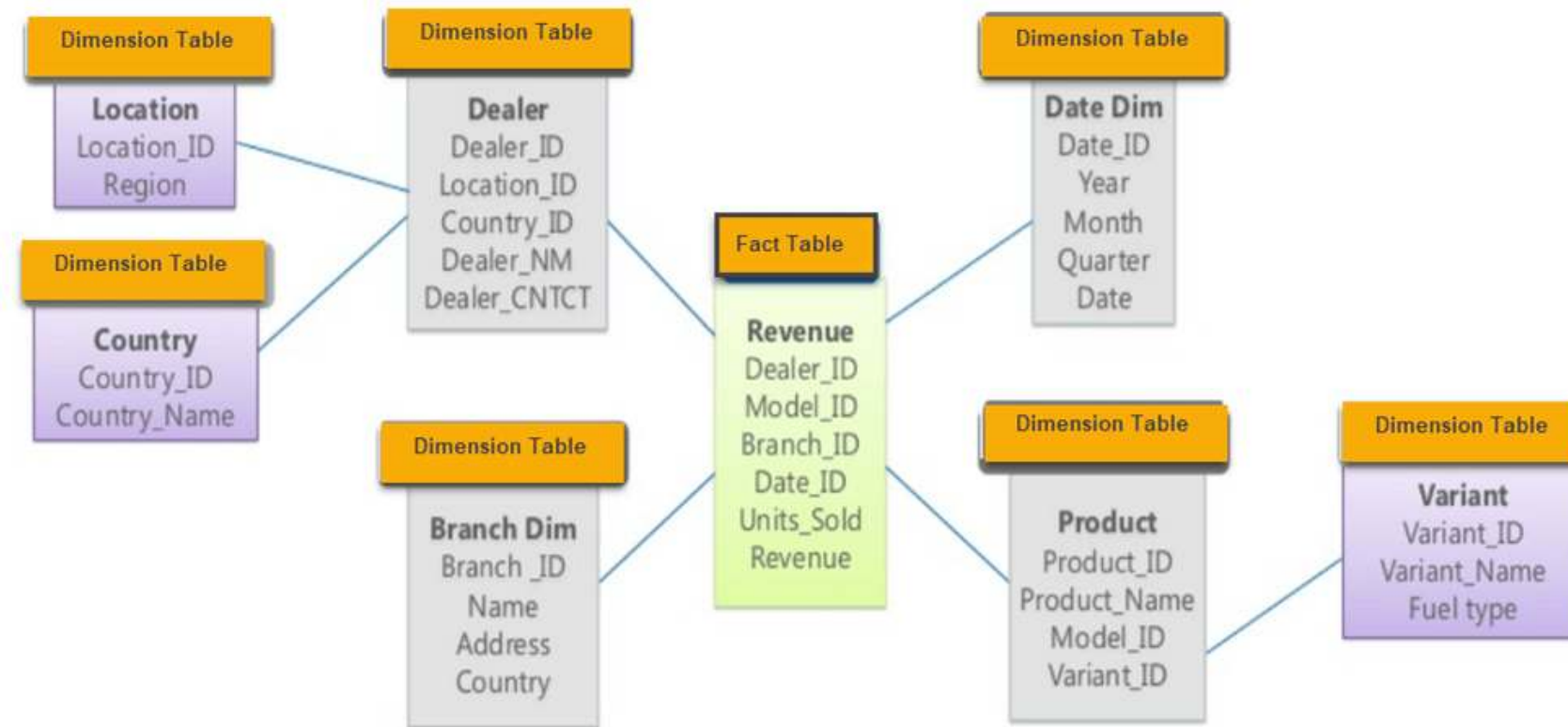
**Note:** FROM Branch f, Product p, Time z, Customer k, Sales v uses **implicit join syntax**. All tables are combined first, and the **join conditions** are defined in the **WHERE** clause (e.g., v.ProductID = p.ProductID). This is equivalent to JOIN ... ON, but separates joins less clearly from filtering conditions.

Note: — Source: Hartung (2011), p. 37

# Snowflake schema



In the Snowflake schema, the redundancies in the Star schema are resolved in the dimension tables and the hierarchy levels are each modeled with their own tables:



— Source: <https://www.guru99.com/star-snowflake-data-warehousing.html>

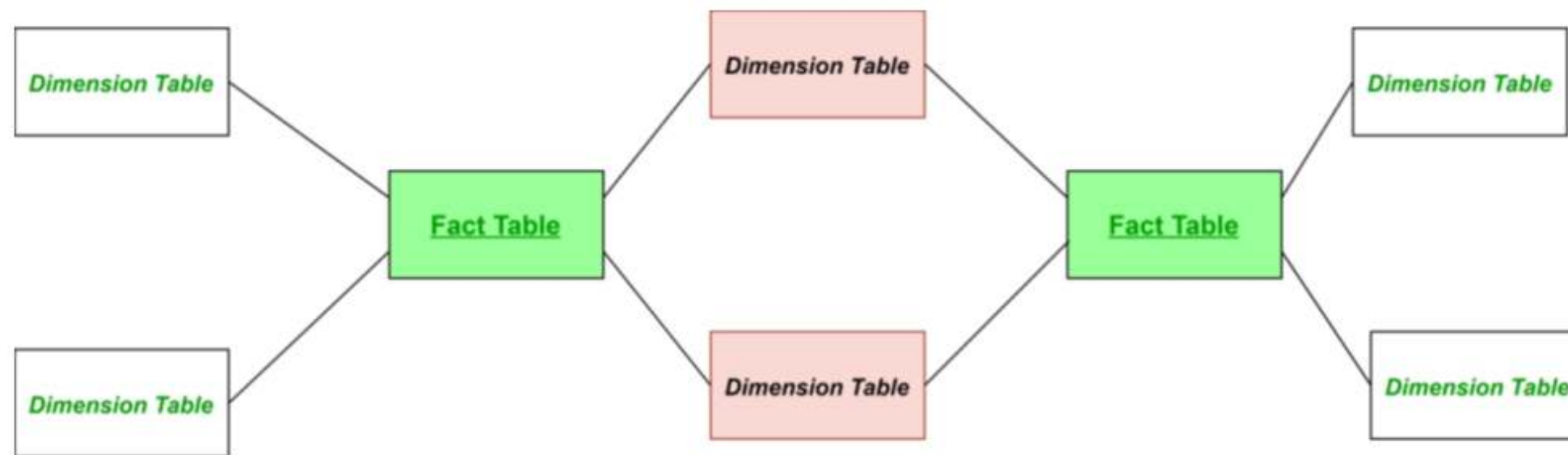
# Star schema vs. snowflake schema

- Speed of query processing is in favor of Star scheme
- Data volume tends to be lower with Snowflake schema
- Snowflake schema is easier to change (maintenance)

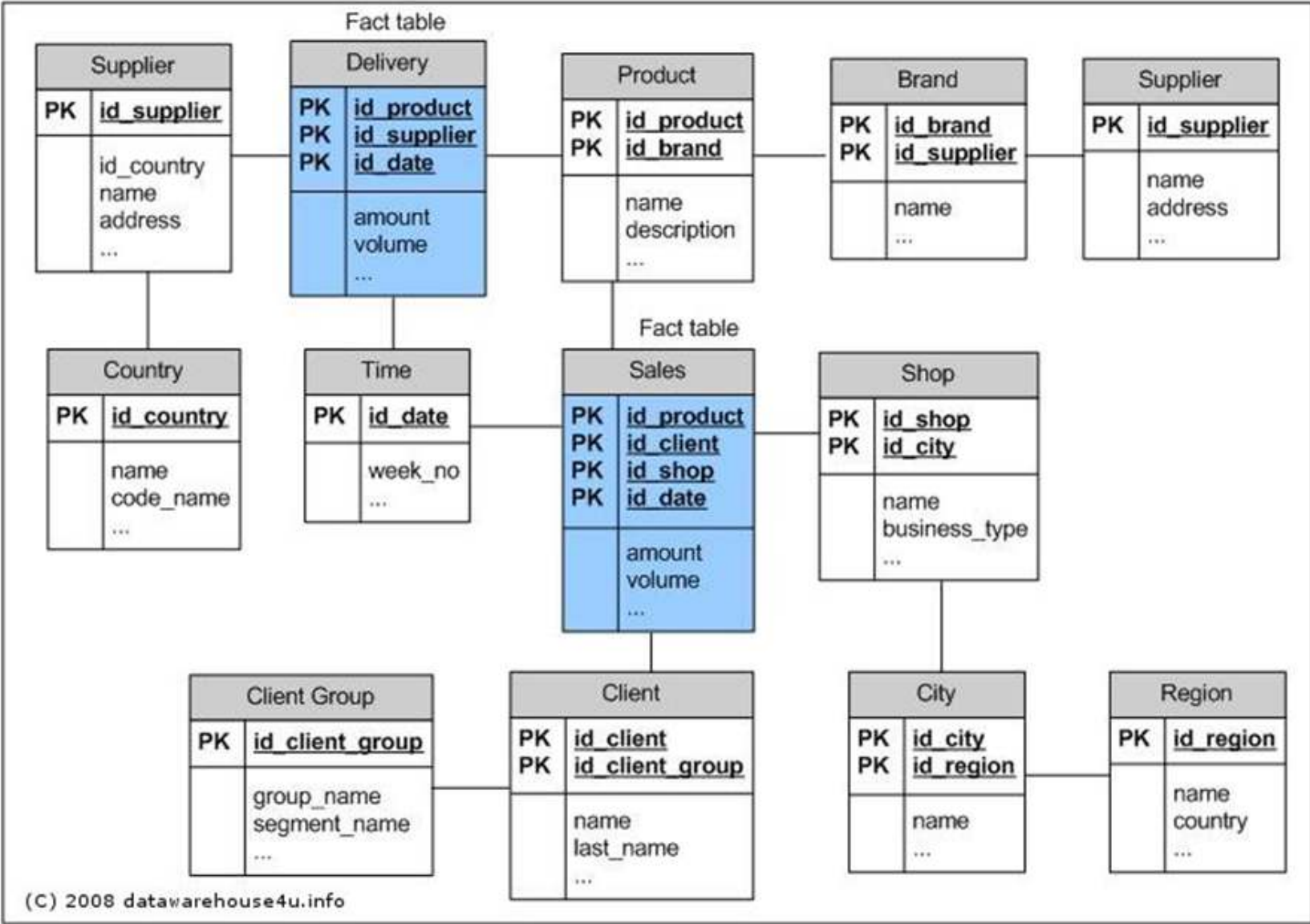
# Galaxy schema



The Galaxy schema contains more than one fact table. The fact tables share some but not all dimensions. The Galaxy schema thus represents more than one data cube (multi-cubes).



# Table structure of the Galaxy schema





# OLAP operations

# Subject of OLAP



Online Analytical Processing (OLAP) is software technology focused on the analysis of dynamic, multidimensional data.

The goal is to provide executives with the information they need quickly, flexibly and interactively.

OLAP functionality can be characterized as dynamic, multidimensional analysis of consolidated enterprise data, which includes the following components:

- Calculations of key figures over different dimensions and aggregation levels
- Trend analysis over definable time intervals
- “What if” analyses
- Multidimensional visualization methods

# Facts and dimensions

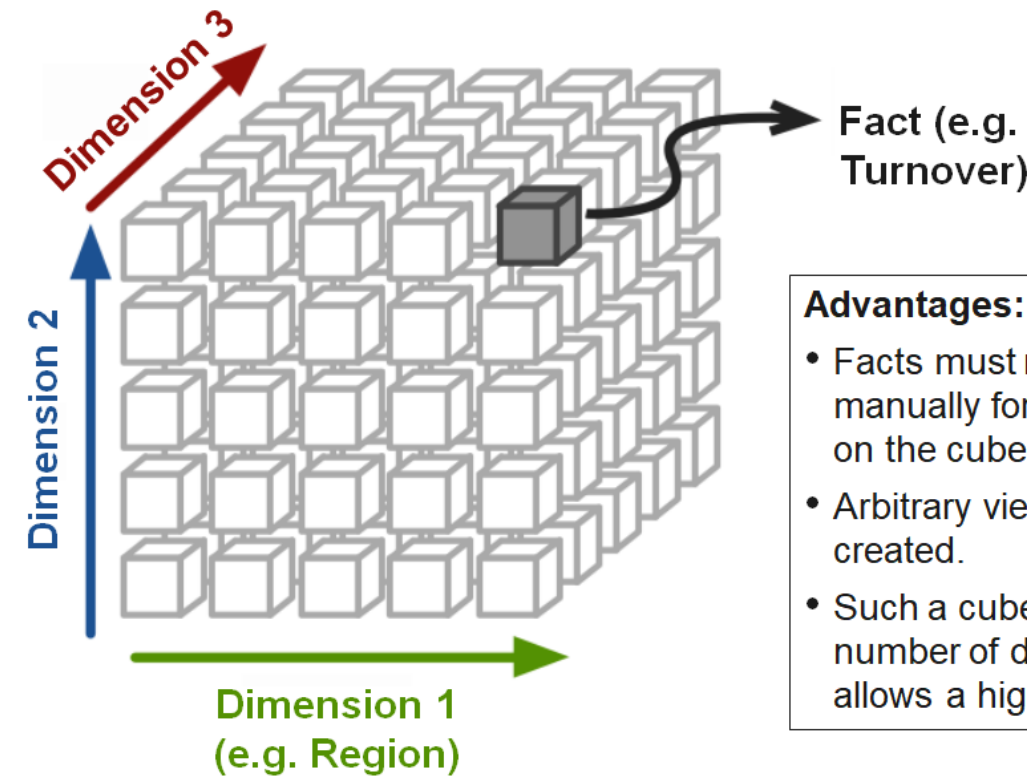


| Turnover      |      | Product Group |        |          |        |         |        |          |        |          |        |
|---------------|------|---------------|--------|----------|--------|---------|--------|----------|--------|----------|--------|
| Region        | Year | Accessories   |        | Desktops |        | Laptops |        | Monitors |        | Printers |        |
|               |      | Actual        | Target | Actual   | Target | Actual  | Target | Actual   | Target | Actual   | Target |
| Austria       | 2019 | 1435          | 1200   | 3444     | 3500   | 3728    | 4100   | 1808     | 1780   | 2001     | 2000   |
|               | 2020 | 3109          | 3145   | 3453     | 3920   | 7558    | 9000   | 1481     | 1610   | 1285     | 1320   |
| Brazil        | 2019 | 2812          | 3000   | 7294     | 7350   | 7529    | 8200   | 3636     | 3550   | 3895     | 3880   |
|               | 2020 | 6079          | 5991   | 6972     | 6720   | 14780   | 15500  | 2952     | 3150   | 2591     | 2640   |
| France        | 2019 | 2910          | 3150   | 7153     | 6650   | 7425    | 8200   | 3549     | 3560   | 4013     | 4000   |
|               | 2020 | 6373          | 6840   | 6594     | 6720   | 15320   | 16300  | 2823     | 2870   | 2541     | 2580   |
| Germany       | 2019 | 4349          | 4500   | 10392    | 10500  | 11355   | 11890  | 5663     | 5630   | 6061     | 6090   |
|               | 2020 | 9370          | 9064   | 10130    | 10080  | 23103   | 24600  | 4290     | 4270   | 3877     | 3780   |
| Great Britain | 2019 | 3619          | 3600   | 9034     | 9100   | 8900    | 8610   | 4440     | 4440   | 4895     | 5020   |
|               | 2020 | 8218          | 7964   | 8432     | 8680   | 19480   | 18900  | 3536     | 3500   | 3227     | 3120   |
| Italy         | 2019 | 2927          | 2850   | 6839     | 6650   | 7534    | 7790   | 3553     | 3550   | 3967     | 4000   |
|               | 2020 | 6215          | 5991   | 6711     | 6720   | 15325   | 16300  | 2852     | 2800   | 2653     | 2700   |
| Switzerland   | 2019 | 2977          | 3150   | 7061     | 6650   | 7551    | 8200   | 3683     | 3710   | 4066     | 4130   |
|               | 2020 | 6163          | 5991   | 6589     | 6720   | 15162   | 14700  | 2837     | 2660   | 2556     | 2520   |
| USA           | 2019 | 14651         | 14700  | 35613    | 35700  | 37595   | 37310  | 18391    | 18380  | 20154    | 20050  |
|               | 2020 | 30900         | 30433  | 32913    | 32760  | 74830   | 73900  | 14137    | 14070  | 12991    | 13020  |

Qualifying Information (Dimensions)

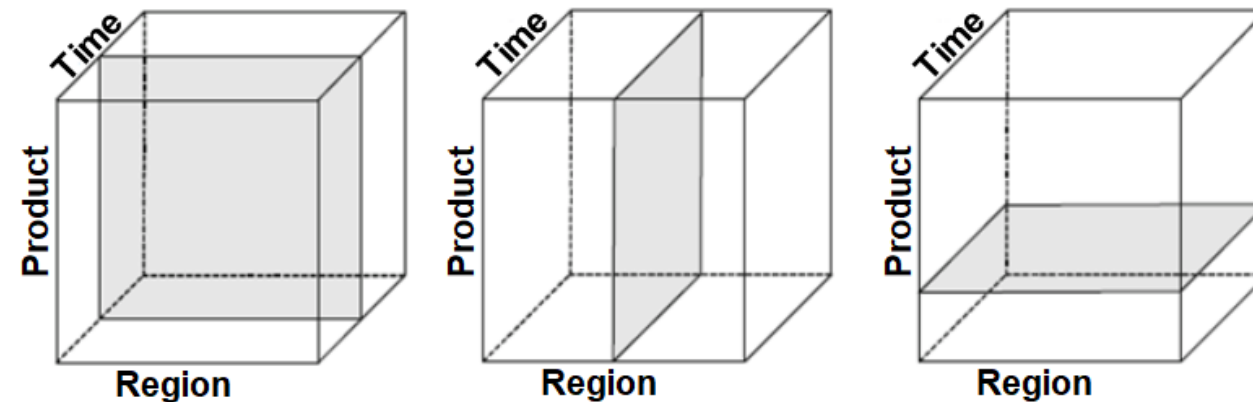
Quantifying Information (Facts)

# Organizing data as hypercube

**Advantages:**

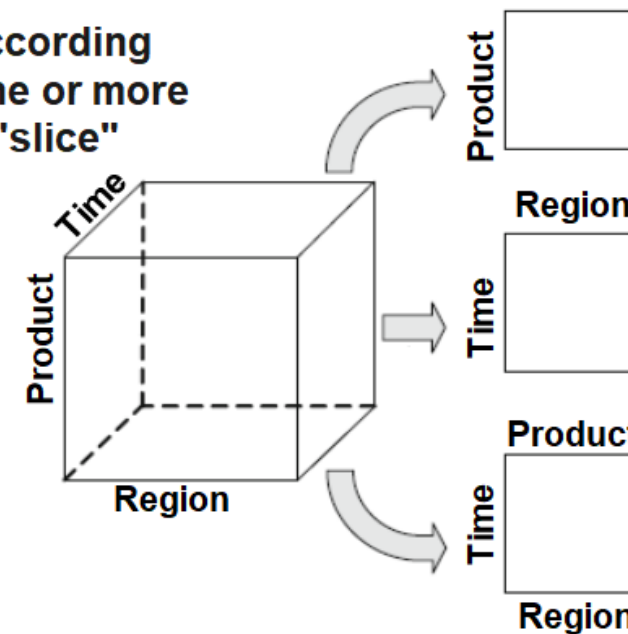
- Facts must not be calculated manually for any chosen view on the cube.
- Arbitrary views can be easily created.
- Such a cube can have any number of dimensions which allows a higher complexity.

# OLAP operations: Slicing and dicing



The **slicing** operator selects tuples from the cube according to the selection condition. The operator specifies one or more values for at least one dimension and thus slices a "slice" from the data cube.

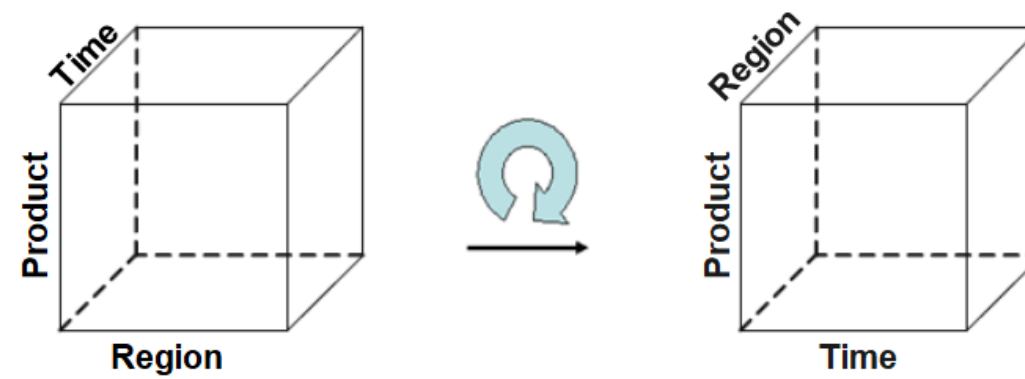
The **dicing** operator selects a subset of the dimensions of the cube and uses it to cut out a smaller "sub-cube".



# OLAP operations: Rotate



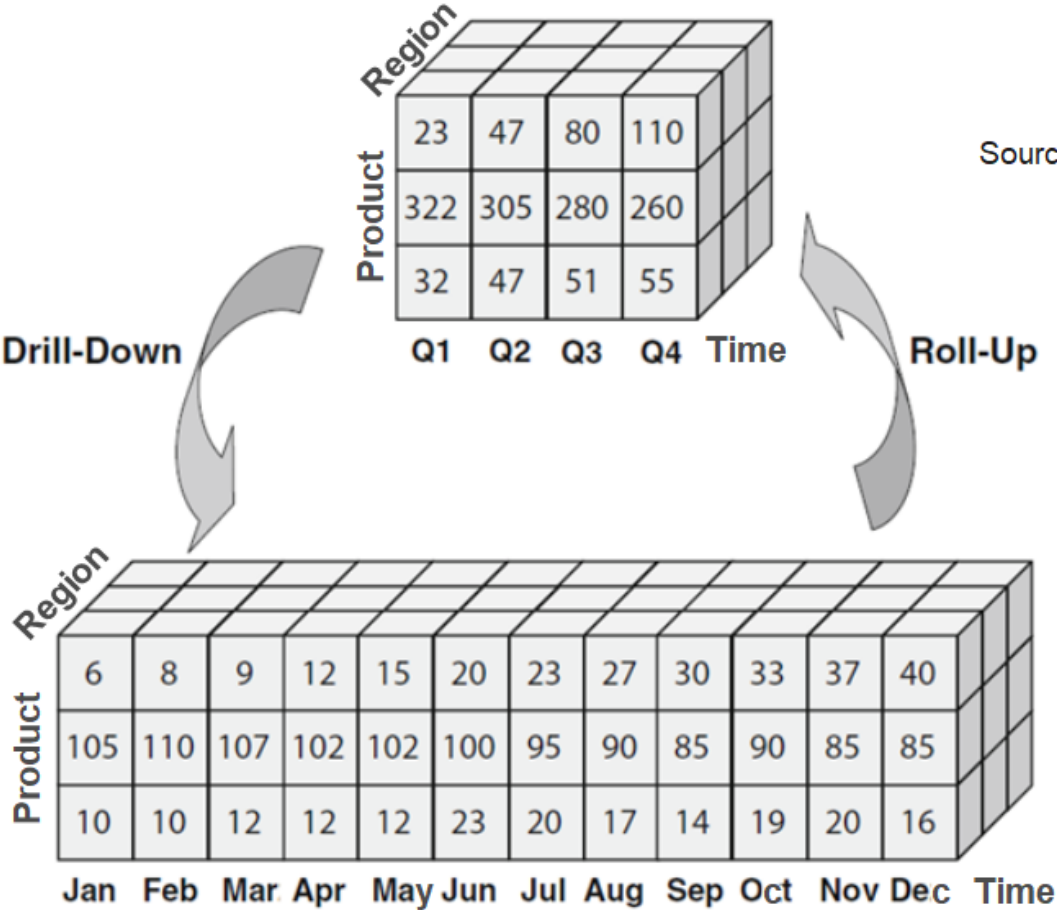
The **rotate** operator rotates the hypercube around one of its axes:



# OLAP operations: Drill down and roll up



The **roll-up** operator corresponds to aggregation and goes from the specific up to the general in the dimensional hierarchy; conversely, **drill-down** goes as disaggregation from the general down to the specific in the hierarchy.



Source: Müller/Lenz (2013), p. 55

# OLAP operations: Nest



The result of a nest operation physically represents a two-dimensional matrix. It is extended by displaying different hierarchy levels of one or more dimensions on one axis (column or row) in nested form. In the following example, three dimensions are displayed:

| Turnover      |      | Product Group |          |         |          |          |        |
|---------------|------|---------------|----------|---------|----------|----------|--------|
| Region        | Year | Accessories   | Desktops | Laptops | Monitors | Printers | Total  |
| Austria       | 2019 | 1435          | 3444     | 3728    | 1808     | 2001     | 12416  |
|               | 2020 | 3109          | 3453     | 7558    | 1481     | 1285     | 16886  |
| Brazil        | 2019 | 2812          | 7294     | 7529    | 3636     | 3895     | 25166  |
|               | 2020 | 6079          | 6972     | 14780   | 2952     | 2591     | 33374  |
| France        | 2019 | 2910          | 7153     | 7425    | 3549     | 4013     | 25050  |
|               | 2020 | 6373          | 6594     | 15320   | 2823     | 2541     | 33651  |
| Germany       | 2019 | 4349          | 10392    | 11355   | 5663     | 6061     | 37820  |
|               | 2020 | 9370          | 10130    | 23103   | 4290     | 3877     | 50770  |
| Great Britain | 2019 | 3619          | 9034     | 8900    | 4440     | 4895     | 30888  |
|               | 2020 | 8218          | 8432     | 19480   | 3536     | 3227     | 42893  |
| Italy         | 2019 | 2927          | 6839     | 7534    | 3553     | 3967     | 24820  |
|               | 2020 | 6215          | 6711     | 15325   | 2852     | 2653     | 33756  |
| Switzerland   | 2019 | 2977          | 7061     | 7551    | 3683     | 4066     | 25338  |
|               | 2020 | 6163          | 6589     | 15162   | 2837     | 2556     | 33307  |
| USA           | 2019 | 14651         | 35613    | 37595   | 18391    | 20154    | 126404 |
|               | 2020 | 30900         | 32913    | 74830   | 14137    | 12991    | 165771 |

# OLAP as Excel plug-in



Microsoft Excel - [Alea Ansicht] - [MIS1]Tabelle1

Datei Bearbeiten Ansicht Einfügen Format Extras Daten Fenster Alea ?

100% 100%

Arial 10

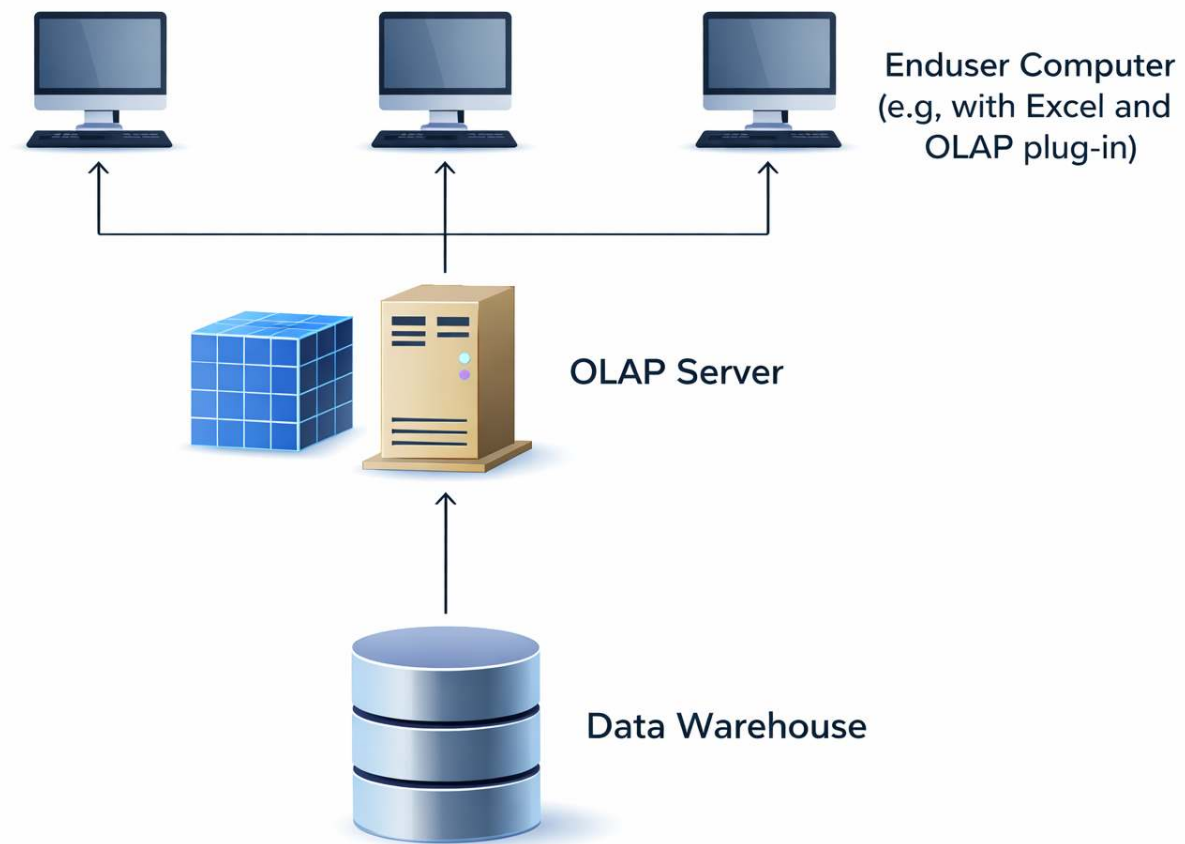
C28 = 386464

|    | B                 | C               | D          | E          | F          | G          | H         | I         | J         |
|----|-------------------|-----------------|------------|------------|------------|------------|-----------|-----------|-----------|
| 19 | LOCAL/TUTOR       |                 |            |            |            |            |           |           |           |
| 20 | VERKAUF           |                 |            |            |            |            |           |           |           |
| 21 | JAHRE             | 1993            |            |            |            |            |           |           |           |
| 22 | DATENART          | Ist             |            |            |            |            |           |           |           |
| 23 | WERTART           | Menge           |            |            |            |            |           |           |           |
| 24 | PRODUKTE          | Desktops gesamt |            |            |            |            |           |           |           |
| 25 |                   |                 |            |            |            |            |           |           |           |
| 26 |                   | MONATE ...      |            |            |            |            |           |           |           |
| 27 | REGIONEN          | Jahr gesamt     | 1. Quartal | 2. Quartal | 3. Quartal | 4. Quartal | Oktober   | November  | Dezember  |
| 28 | Welt gesamt       | 386.464,00      | 102.755,00 | 101.392,00 | 70.983,00  | 111.334,00 | 36.588,00 | 37.386,00 | 37.360,00 |
| 29 | Europäische Union | 150.483,00      | 39.720,00  | 39.721,00  | 27.616,00  | 43.426,00  | 14.108,00 | 14.782,00 | 14.536,00 |
| 30 | Österreich        | 7.264,00        | 1.941,00   | 1.864,00   | 1.320,00   | 2.139,00   | 724,00    | 734,00    | 681,00    |
| 31 | Belgien           | 7.263,00        | 1.871,00   | 1.902,00   | 1.374,00   | 2.116,00   | 664,00    | 734,00    | 718,00    |
| 32 | Dänemark          | 7.213,00        | 1.866,00   | 1.876,00   | 1.339,00   | 2.132,00   | 706,00    | 702,00    | 724,00    |
| 33 | Frankreich        | 14.648,00       | 3.836,00   | 3.866,00   | 2.665,00   | 4.281,00   | 1.447,00  | 1.452,00  | 1.382,00  |
| 34 | Deutschland       | 21.721,00       | 5.769,00   | 5.739,00   | 4.002,00   | 6.211,00   | 2.077,00  | 2.101,00  | 2.033,00  |
| 35 | Großbritannien    | 18.000,00       | 4.802,00   | 4.810,00   | 3.232,00   | 5.156,00   | 1.642,00  | 1.811,00  | 1.703,00  |
| 36 | Griechenland      | 7.031,00        | 1.971,00   | 1.900,00   | 1.338,00   | 1.822,00   | 599,00    | 621,00    | 602,00    |
| 37 | Irland            | 5.514,00        | 1.444,00   | 1.473,00   | 989,00     | 1.608,00   | 507,00    | 568,00    | 533,00    |
| 38 | Luxemburg         | 3.710,00        | 981,00     | 943,00     | 678,00     | 1.108,00   | 360,00    | 378,00    | 370,00    |
| 39 | Niederlande       | 14.857,00       | 3.949,00   | 3.858,00   | 2.642,00   | 4.408,00   | 1.352,00  | 1.531,00  | 1.525,00  |
| 40 | Portugal          | 6.934,00        | 1.820,00   | 1.912,00   | 1.363,00   | 1.839,00   | 605,00    | 621,00    | 613,00    |
| 41 | Spanien           | 21.914,00       | 5.738,00   | 5.850,00   | 3.953,00   | 6.373,00   | 2.042,00  | 2.123,00  | 2.208,00  |
| 42 | Italien           | 14.414,00       | 3.732,00   | 3.728,00   | 2.721,00   | 4.233,00   | 1.383,00  | 1.406,00  | 1.444,00  |
| 43 | Europa (OST/WEST) | 247.770,00      | 65.812,00  | 65.527,00  | 45.787,00  | 70.644,00  | 23.254,00 | 23.828,00 | 23.562,00 |
| 44 |                   |                 |            |            |            |            |           |           |           |

Tabelle1

Bereit

# Architecture of an OLAP system



# Data warehousing in modern analytics

## DWH as foundation

- Codifies data preparation (ETL), transformations, and integration
- Centralizes **data access** and **metadata knowledge**

## Evolution vs. r eplacement

- Accumulated logic and data understanding **favor incremental over greenfield evolution**

## Data preparation

- Errors should be fixed **at the source systems**
- Transformations handled in **ETL processes**
- Duplicate handling depends on context (within vs. across datasets)

## Role in modern analytics

- Serves as a **curated, reliable data layer**
- ML and advanced analytics typically operate **on top of the DWH**

# Summary



- A **data warehouse** supports analytical decision-making by integrating data from multiple sources. It is **subject-oriented, integrated, time-variant, and non-volatile**, and separates analytical workloads from operational systems.
- The **ETL process** (Extract, Transform, Load) prepares data for analysis: data is extracted from sources, cleaned and integrated in a staging area, and loaded into the warehouse. **Metadata** ensures transparency and usability.
- **Dimensional modeling** organizes data for analysis:
  - **Facts** (e.g., sales) and **dimensions** (e.g., time, product, region)
  - Common schemata: **star** (simple, fast), **snowflake** (normalized), **galaxy** (multiple fact tables)
- **OLAP systems** enable interactive, multidimensional analysis: operations such as **slice, dice, drill-down, and roll-up** support flexible exploration of aggregated data.
- Overall, analytical data architecture transforms **operational data into structured information that can be efficiently queried** to support reporting, dashboards, and decision-making.

# Survey: Session 3



<https://forms.gle/dkwMZN4zPT7N3qbx5>

Note: Responses may be analyzed and published in anonymized form.

Please complete the survey before you leave today — thank you 🙏



# References

Inmon, W. H. (1995). What is a data warehouse? *Prism Tech Topic*, 1(1), 1–5.

Inmon, W. H. (2005). *Building the data warehouse*. John wiley & sons.